a Roll the camera

Wall moves; lens shading stays.

Radial shading is unchanged by rolls.

b Separate lens and scene

SENSOR FIELD V(x)

Physical base + neural residual

Radial envelope + bounded spatial detail

SCENE FIELD L(u)

Polynomial + neural residual

Evaluated in the moving scene frame

Jointly fit pose and exposure

c Select radial shading

Focal length + sensor size

Reduce broad illumination leakage

Mix fitted + analytic radial profiles

Retain the spatial remainder

Shading map → later image correction

ROLLS CONSTRAIN SPATIAL STRUCTURE • METADATA SELECTS THE RADIAL TREND